

DAILY CURRENT AFFAIRS AUTOMATION
System Architecture, Prompts Engineering & Operational Manual
Kerala PSC • SSC • UPSC • Banking | Malayalam Presentation Engine

1. Executive Summary & 6:00 AM Workflow

Purpose & Objective

This automated system transforms daily news from **The Hindu** and official government sources into **exam-curated Malayalam presentation slide decks and PDFs** for your daily YouTube educational broadcasts.

The 6:00 AM Morning Routine

When you wake up at **6:00 AM**, you run one command in your terminal:



Total Execution Time: ~10 to 15 seconds

In 10 seconds, the system fetches today's morning paper news, extracts photos, queries Google Gemini, formats Malayalam text, and automatically opens your Presentation PDF in Apple Preview.

Core Deliverables Generated Daily:

1. **10 to 15 Exam Stories** categorized into **Kerala, National (India), Sports, and International**.
2. **2 High-Yield Exam Vocabulary Words** with English/Malayalam definitions & example sentences.
3. **Exact Newspaper Photos** downloaded and displayed on each slide.
4. **Print & Screen-Ready PDF** formatted in 16:9 widescreen layout.

2. End-to-End System Architecture



3. Data Ingestion: Official News Feeds

Instead of slow, fragile webpage scraping that can get blocked, the system connects directly to **official, public RSS feeds**:

Category	Official Source	Endpoint URL
National (India)	The Hindu National Desk	<code>https://www.thehindu.com/news/national/feeder/default.rss</code>
Kerala State	The Hindu Kerala Bureau	<code>https://www.thehindu.com/news/national/kerala/feeder/default.rss</code>
Business & Economy	The Hindu Business Desk	<code>https://www.thehindu.com/business/feeder/default.rss</code>
Sports	The Hindu Sports Desk	<code>https://www.thehindu.com/sport/feeder/default.rss</code>
International	The Hindu World Desk	<code>https://www.thehindu.com/news/international/feeder/default.rss</code>
Kerala Government	PRD Kerala (Official PR)	<code>https://prd.kerala.gov.in/en/rss.xml</code>

Why This Method is Best:

- **100% Legal & Safe:** Uses public syndicate endpoints provided by the publisher (Kasturi & Sons Ltd.).
- **Ultra Fast:** Ingests 300+ daily articles in less than 1.5 seconds.
- **Contains Official Photos:** Each RSS item includes the `<media:content>` tag with the high-resolution photo from the print article.

4. Master Prompt Suite: Overview & Design

The AI curation is governed by 6 specialized prompts located in the `prompts/` directory:

Prompt File	Role & Objective	Target Output
[<code>prompts/01_news_selector.md</code>] (file:///Users/pious/Documents/personalproject/scrapping/prompts/01_news_selector.md)	Chief Editor: Filters 200+ raw news down to 10–15 high-yield exam stories.	Filtered JSON list with category & exam relevance.
[<code>prompts/02_fact_researcher.md</code>] (file:///Users/pious/Documents/personalproject/scrapping/prompts/02_fact_researcher.md)	Fact Verifier: Extracts dates, statutory acts, ministries, and confidence scores (≥ 0.85).	Verified facts and cross-check notes.
[<code>prompts/03_malayalam_writer.md</code>] (file:///Users/pious/Documents/personalproject/scrapping/prompts/03_malayalam_writer.md)	Malayalam Educator: Translates & formats news into presentation slides.	Malayalam headline, sub-headline, summary, points.
[<code>prompts/04_mcq_generator.md</code>] (file:///Users/pious/Documents/personalproject/scrapping/prompts/04_mcq_generator.md)	Quiz Master: Generates PSC/UPSC standard 4-option multiple-choice questions.	4-option MCQs with correct answers & explanations.
[<code>prompts/05_vocabulary_generator.md</code>] (file:///Users/pious/Documents/personalproject/scrapping/prompts/05_vocabulary_generator.md)	Language Specialist: Selects exactly 2 high-yield English vocabulary words.	Word, part of speech, Eng/Mal definitions, example.
[<code>prompts/06_youtube_script.md</code>] (file:///Users/pious/Documents/personalproject/scrapping/prompts/06_youtube_script.md)	YouTube Scriptwriter: Creates natural Malayalam narration for voice-over recording.	Timestamped conversational voice-over script.

5. Deep-Dive: News Selector & Malayalam Writer Prompts

Prompt 01: AI News Selector ([prompts/01_news_selector.md]

(file:///Users/pious/Documents/personalproject/scrapping/prompts/01_news_selector.md))

- **Editor Persona:** Specialized in Kerala PSC (Degree & 10th Level), SSC CGL, UPSC Prelims, and Banking exams.
- **Strict 4-Way Categorization:**
 - i. **Kerala Affairs** : State government policies, infrastructure, gazettes, PRD announcements.
 - ii. **India / National & Economy** : Union Cabinet, Supreme Court, RBI, economic policies, science & tech.
 - iii. **Sports** : Major championships, Olympic sports, national records, award recipients.
 - iv. **International** : Bilateral treaties, global summits (G20, ASEAN, BRICS), UN resolutions.
- **Strict Exclusions:** Celebrity gossip, minor crime, sensationalism, political rallies, clickbait.

Prompt 03: Malayalam Content & Slide Writer ([prompts/03_malayalam_writer.md]
(file:///Users/pious/Documents/personalproject/scrapping/prompts/03_malayalam_writer.md))

- Linguistic Guidelines:

- Write **natural, conversational Malayalam** (avoid robotic, word-for-word machine translation).
- Keep proper nouns, institutions, and technical terms in English or standard transliteration (e.g. *'EXIM Services'*, *'Repo Rate'*, *'MeitY'*, *'PPP മാതൃക'*).
- Structure bullet points for Kerala PSC factual recall (Nodal agencies, target years, allocations).

6. Deep-Dive: Vocabulary & MCQ Prompts

Prompt 05: Daily Vocabulary Builder ([prompts/05_vocabulary_generator.md]
(file:///Users/pious/Documents/personalproject/scrapping/prompts/05_vocabulary_generator.md))

- **Goal:** Enhances English Verbal Ability for SSC CGL, Banking (IBPS/SBI), and Kerala PSC English.
- **Selection Rule:** Exactly 2 high-yield, sophisticated English words found in today's news reporting (e.g. *Indigenous, Pragmatic, Moratorium, Exacerbate*).
- **Format:**
 - Word in UPPERCASE + Part of Speech (Adjective / Noun / Verb).
 - Concise English definition + Precise Malayalam meaning (സ്വദേശീയമായ , പ്രായോഗികമായ).
 - Illustrative sentence contextualized with today's news events.

**Prompt 04: MCQ & Quiz Generator ([prompts/04_mcq_generator.md]
(file:///Users/pious/Documents/personalproject/scrapping/prompts/04_mcq_generator.md))**

- **Exam Relevance:** Follows Kerala PSC Statement-based & factual question standards.
- **Format:**
 - Clear Malayalam question stem.
 - 4 distinct options (A, B, C, D) with realistic distractors.
 - Detailed explanation in Malayalam detailing why the correct answer is right.

7. How 6:00 AM Morning Filtering Works

The Challenge of Morning News

A physical morning newspaper delivered at 6:00 AM contains news finalized over the **previous 24 hours** (yesterday morning through late night). A simple calendar check (`date == today`) would return almost no news at 6:00 AM because journalists are asleep between midnight and dawn.

The Solution: 24-Hour Rolling Window

```
const      new Date    // 6:00 AM today
const      getTime     24  60  60  1000  // Yesterday 6:00 AM

// Filters articles strictly within the print edition cycle
if          getTime
  continue // Discard older articles
```

What this guarantees:

Whether you run the script at **6:00 AM, 7:00 AM, 8:00 AM, or 9:00 AM**, you will always capture 100% of the stories printed in today's morning paper.

8. Why is it 100% Free?

Google AI Studio Free Tier Quota

Google provides free access to Gemini Flash models under the following terms:

Metric	Free Tier Quota	Your Usage
Daily Requests	1,500 requests / day	1 request / day (at 6:00 AM)
Rate Limit	15 requests / minute	1 request
Billing Required	NO (No credit card attached)	Free forever
Quota Consumption	100% available	0.06% daily usage

 Zero Financial Cost:

Since the entire daily curation is bundled into 1 smart JSON prompt, you only make 1 API request per day. You will never be charged.

9. Slide Layout & Marp PDF Engine

Visual Layout (Matching YouTube Educational Format)

- **16:9 Aspect Ratio** (1280px × 720px).
- **Left Column (55%):**
 - Category Badge (🌴 KERALA AFFAIRS , 🏛️ INDIA AFFAIRS , 🏆 SPORTS , 🌐 INTERNATIONAL).
 - Bold Navy Malayalam Headline.
 - Blue Sub-headline highlighting the core development.
 - 2-3 sentence Malayalam summary.
 - Mint-green fact box: 📍 പ്രധാന വസ്തുതകൾ & പരീക്ഷാ പോയിന്റുകൾ .
- **Right Column (45%):**
 - High-resolution lead photograph downloaded directly from the news article.
- **Full-Width Fallback (100%):**
 - If an article has no photo, the card automatically expands to full width with zero empty space.

PDF Compilation Command

The script uses `@marp-team/marp-cli` to render the styled layout into a vector PDF:



10. Complete Project File Guide

File Path	Description
[scripts/fetch_and_generate_today.js] (file:///Users/pious/Documents/personalproject/scrapping/scripts/fetch_and_generate_today.js)	Main Engine: Ingestion, 24-hr filter, Gemini AI, photo downloader & PDF generator.
[scripts/generate_pdf.sh](file:///Users/pious/Documents/personalproject/scrapping/scripts/generate_pdf.sh)	Standalone script to compile any report Markdown file into a PDF.
[scripts/generate_slides.sh](file:///Users/pious/Documents/personalproject/scrapping/scripts/generate_slides.sh)	Converts reports into interactive HTML slides for browser presentations.
[config/sources.json](file:///Users/pious/Documents/personalproject/scrapping/config/sources.json)	Central catalog of all primary and secondary RSS feeds.
[config/default_config.json](file:///Users/pious/Documents/personalproject/scrapping/config/default_config.json)	Configuration (10-20 stories, 2 vocabulary words, exam targets).
[prompts/](file:///Users/pious/Documents/personalproject/scrapping/prompts/)	Master prompt templates (News selector, Malayalam writer, Vocabulary, MCQs).
[reports/](file:///Users/pious/Documents/personalproject/scrapping/reports/)	Daily <code>.md</code> reports and compiled <code>_Presentation.pdf</code> files.
[reports/images/](file:///Users/pious/Documents/personalproject/scrapping/reports/images/)	Downloaded high-resolution newspaper article photos.
[workflows/](file:///Users/pious/Documents/personalproject/scrapping/workflows/)	Visual n8n workflow blueprints.

11. Step-by-Step User Guide & Automation

Daily Manual Execution (Recommended)

1. Open Terminal on your Mac.
2. Navigate to your project folder:

```
cd
```

3. Run the pipeline:

4. Your presentation PDF will pop open automatically in **Apple Preview** in ~10 seconds!

Optional: 100% Hands-Free Mac Cron (Automatic 6:00 AM Trigger)

If you want the PDF to be ready automatically when you open your laptop:

1. In Terminal, run: `crontab -e`
2. Add this line and save:

```
cd
```

- Every morning at 6:00 AM, the PDF will be generated and waiting for you!